



Faculty of Computer Science and Mathematics
Universiti Malaysia Terengganu

LAB REPORT

CSE3023 Web-Based Application Development

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Lab Report	1
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Exercise:

Full source code:

1. Correct creation of Servlet (GreetingServlet)

GreetingServlet.java
<pre>import java.io.IOException; import java.io.PrintWriter; import javax.servlet.ServletException; import javax.servlet.http.HttpServlet; import javax.servlet.http.HttpServletRequest; import javax.servlet.http.HttpServletResponse; import javax.servlet.RequestDispatcher; /** * * @author wpy92 */ public class GreetingServlet extends HttpServlet { /** * Processes requests for both HTTP <code>GET</code> and * <code>POST</code> * methods. * * @param request servlet request * @param response servlet response * @throws ServletException if a servlet-specific error occurs * @throws IOException if an I/O error occurs */ protected void processRequest(HttpServletRequest request, HttpServletResponse response) throws ServletException, IOException { response.setContentType("text/html;charset=UTF-8"); try (PrintWriter out = response.getWriter()) { /* TODO output your page here. You may use following sample code. */ out.println("<!DOCTYPE html>"); out.println("<html>"); out.println("<head>"); out.println("<title>Servlet GreetingServlet</title>"); out.println("</head>"); out.println("<body>"); out.println("<h1>Servlet GreetingServlet at " + request.getContextPath() + "</h1>"); } } }</pre>

```

        out.println("</body>");
        out.println("</html>");
    }
}

// <editor-fold defaultstate="collapsed" desc="HttpServlet
methods. Click on the + sign on the left to edit the code.">
/**
 * Handles the HTTP <code>GET</code> method.
 *
 * @param request servlet request
 * @param response servlet response
 * @throws ServletException if a servlet-specific error occurs
 * @throws IOException if an I/O error occurs
 */
@Override
protected void doGet(HttpServletRequest request,
HttpServletResponse response)
    throws ServletException, IOException {

    // Get parameter "name"
    String name = request.getParameter("name");

    // Check if null or empty
    if(name == null || name.isEmpty()) {
        name = "Guest";
    }

    // Send data to JSP
    request.setAttribute("username", name);

    // Forward to JSP
    RequestDispatcher dispatcher =
request.getRequestDispatcher("greeting.jsp");
    dispatcher.forward(request, response);
}

/**
 * Handles the HTTP <code>POST</code> method.
 *
 * @param request servlet request
 * @param response servlet response
 * @throws ServletException if a servlet-specific error occurs
 * @throws IOException if an I/O error occurs
 */
@Override

```

```
        protected void doPost(HttpServletRequest request,
        HttpServletResponse response)
            throws ServletException, IOException {
            processRequest(request, response);
        }

        /**
         * Returns a short description of the servlet.
         *
         * @return a String containing servlet description
         */
        @Override
        public String getServletInfo() {
            return "Short description";
        } // </editor-fold>
    }
```

greeting.jsp

```
<%--
    Document      : greeting.jsp
    Created on   : Apr 2, 2026, 4:59:31 AM
    Author      : wpy92
--%>

<%@ page import="java.util.Date" %>

<%@page contentType="text/html" pageEncoding="UTF-8"%>
<!DOCTYPE html>
<html>
    <head>
        <meta http-equiv="Content-Type" content="text/html;
charset=UTF-8">
        <title>Greeting Page</title>
    </head>
    <body>
        <h1>Hello, <%= request.getAttribute("username") %>!</h1>

        <p>
            <%
                Date currentDate = new Date();
                out.println("<p>Current date and time is " +
currentDate.toString() + "</p>");
            %>

            <% response.setIntHeader("Refresh", 5); %>
        </p>
    </body>
</html>
```

2. Correct handling of request parameter (name) with default value (Guest)

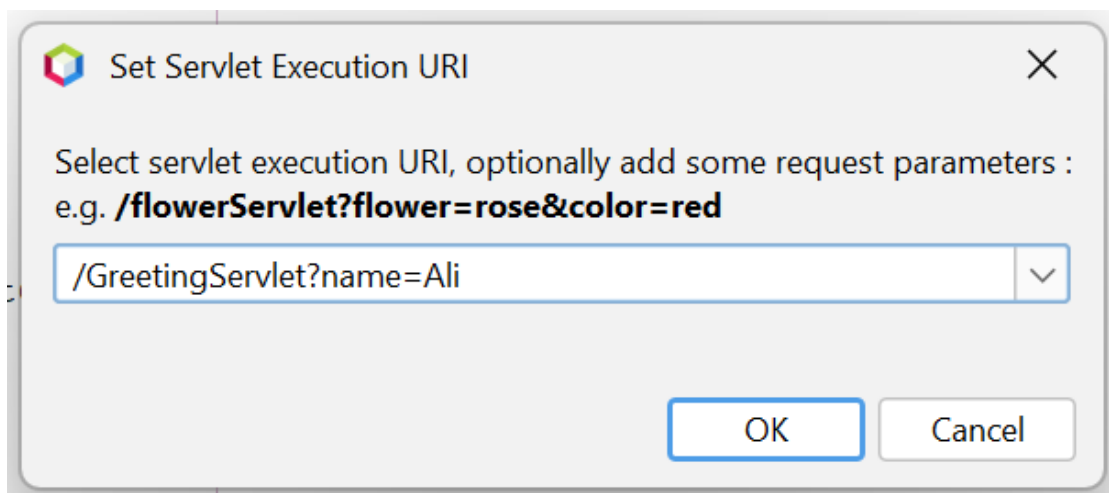
Source code:

```
// Get parameter "name"
String name = request.getParameter("name");

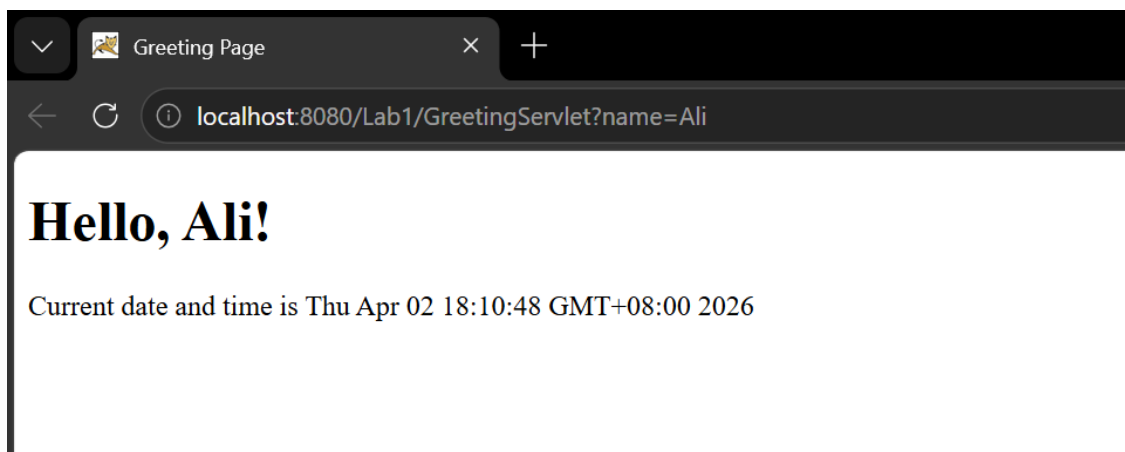
// Check if null or empty
if(name == null || name.isEmpty()) {
    name = "Guest";
}
```

Result 1

If name entered:

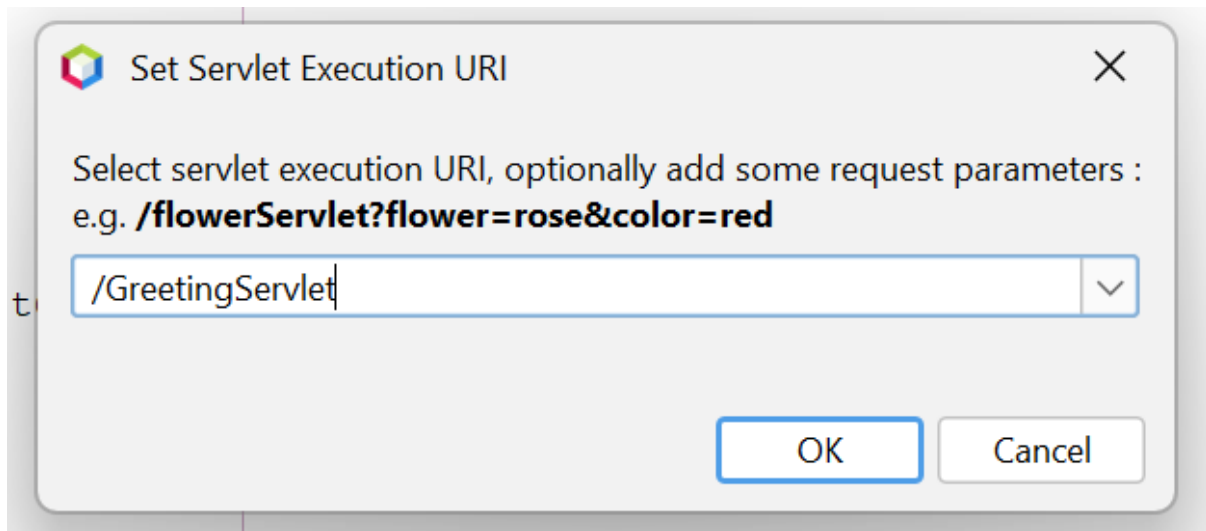


Output:

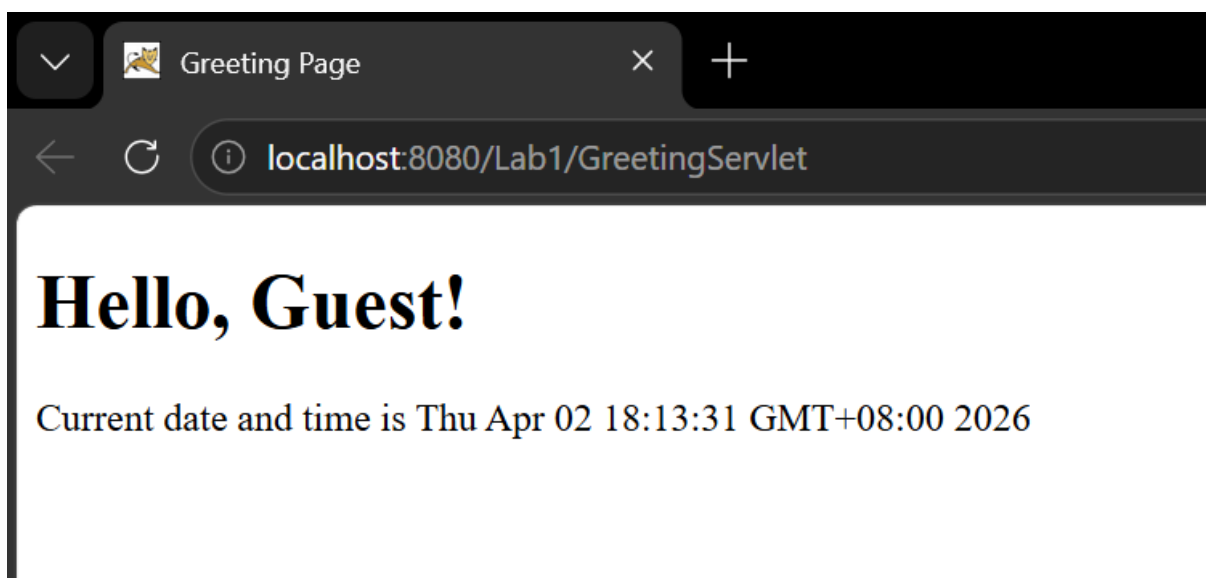


Result 2:

If name is not entered:



Output:



3. Successful forwarding from Servlet to JSP (greeting.jsp)

```
// Forward to JSP
RequestDispatcher dispatcher = request.getRequestDispatcher("greeting.jsp");
dispatcher.forward(request, response);
```

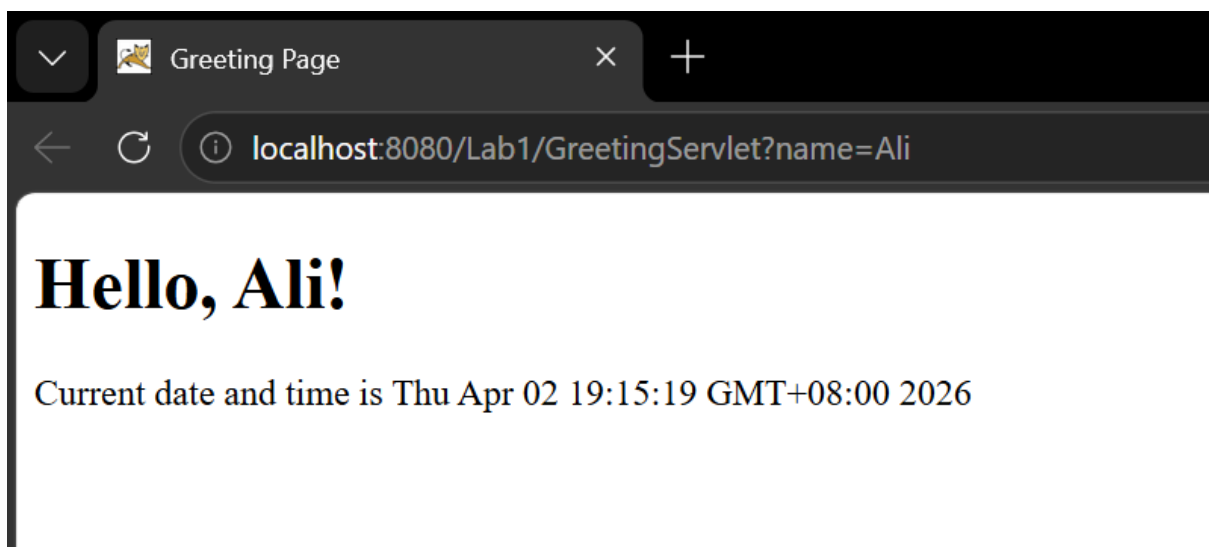
4. Correct display of dynamic content (name and current date/time) in JSP

Source code:

```
<h1>Hello, <%= request.getAttribute("username") %>!</h1>

<p>
<%
    Date currentDate = new Date();
    out.println("<p>Current date and time is " + currentDate.toString() + "</p>");
%>
```

Output:



5. Implementation of auto-refresh functionality (every 5 seconds)

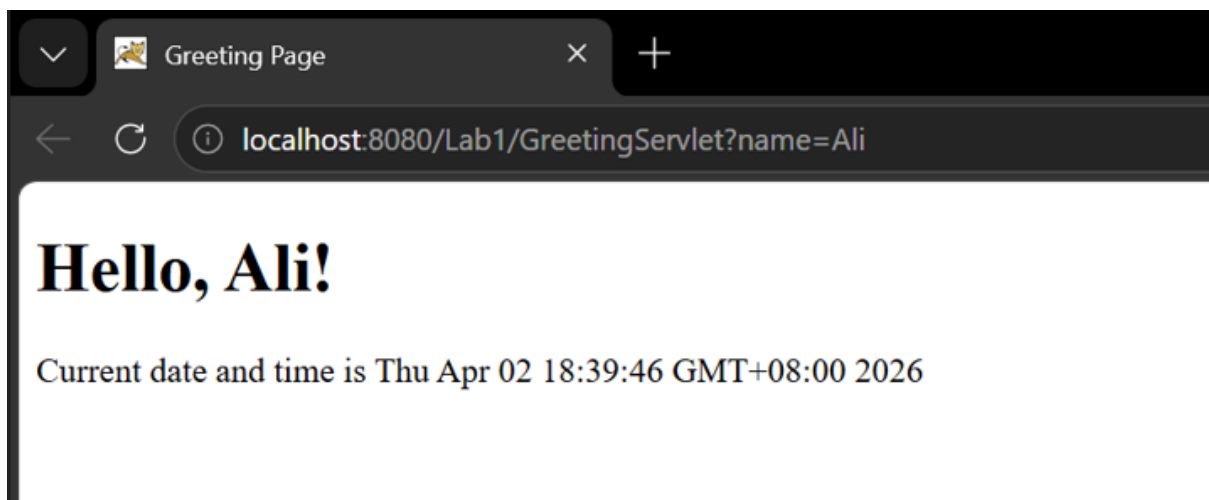
Source code:

```
<% response.setHeader("Refresh", 5); %>
```

Result 3:

The date and time will update automatically every 5 seconds.

Output:



→

